

Name: _____

Maths Class: _____

SYDNEY TECHNICAL HIGH SCHOOL



TRIAL HIGHER SCHOOL CERTIFICATE

2008

EXTENSION 1 MATHEMATICS

Instructions:

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question
- Start each question on a new page

Total Marks – 84

- Attempt Questions 1 – 7
- All questions are of equal value

(For markers use only)

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Total

Question 1

- a) Differentiate:
- i) $x^2 \cos^{-1} x$ 2
 - ii) $\log_{10} 3x$ 2
- b) There is a remainder of 1 when $P(x) = x^3 - 3x^2 + px - 14$ is divided by $x - 3$. Find the value of p . 2
- c) Find the simultaneous solution of: $|x - 3| < 4$ and $|x - 1| > 1$ 3
- d) The point $P(3, 5)$ divides the interval joining $A(-1, 1)$ and $B(5, 7)$ internally in the ratio $m:n$. Find $m:n$. 2
- e) Find $\int \cos x \sin x \, dx$ 1

Question 2 (Start a new page)

- a) Find $\lim_{x \rightarrow \infty} \frac{3x^2 - 7x}{5 + x^2}$ 1
- b) Find the acute angle, to the nearest degree, between the curve $y = x^2$ and the line $5x - y - 6 = 0$ at the point of intersection $(3, 9)$ 2
- c) i) Solve $t^2 + 2t - 1 = 0$ 1
- ii) Using your results from part i), and the expansion for $\tan 2\theta$, find the exact value of $\tan 22.5^\circ$. Simplify your answer. 2

- d) i) Express $3 \cos x - 2 \sin x$ in the form $A \cos(x + \alpha)$ where $A > 0$ and $0^\circ \leq \alpha \leq 90^\circ$ 2
- ii) Hence find the smallest positive x degrees such that $3 \cos x - 2 \sin x$ has a maximum value (do not use calculus). Give your answer correct to 1 decimal place. 1
- e) Express $\sin(\tan^{-1}x + \tan^{-1}y)$ in terms of x and y only. 3

Question 3 (Start a new page)

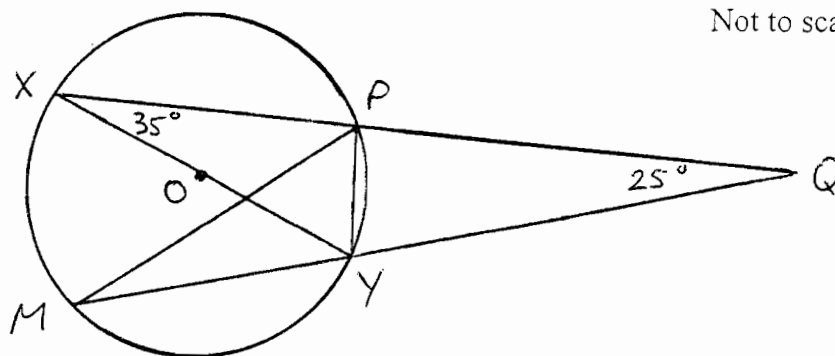
- a) Solve for $0 \leq \theta \leq 2\pi$: $\cos 2\theta = \cos^2 \theta$ 2
- b) Solve $\frac{x^2}{x-4} < 0$ 2
- c) Find $\int \frac{x+4}{x^2+4} dx$ 2
- d) Use the substitution $u = e^x$ to find $\int \frac{e^x}{\sqrt{9-4e^{2x}}} dx$ 3
- e) α, β, γ are the roots of the equation $2x^3 + 5x - 3 = 0$ 3
- Find the value of $\alpha^2 + \beta^2 + \gamma^2$.

Question 4 (Start a new page)

a)

Not to scale

3



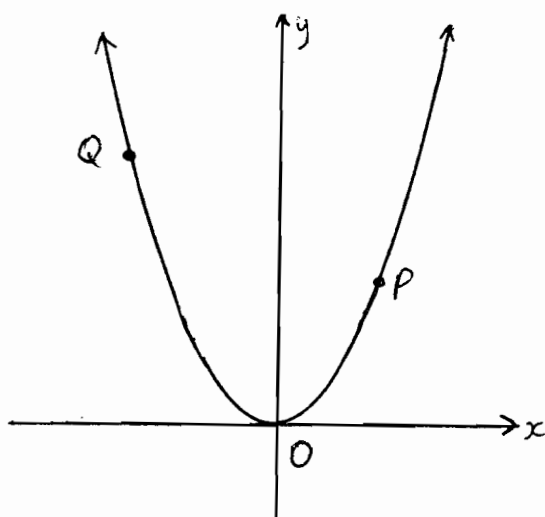
O is the centre of the circle

$$\angle PXY = 35^\circ \text{ and } \angle PQY = 25^\circ$$

i) Copy the diagram onto your answer paper

ii) Find $\angle MPY$ giving full reasons

b)



The points $P(2p, p^2)$ and $Q(2q, q^2)$

move on the parabola $x^2 = 4y$ such that the chord PQ subtends a right angle at the origin O

i) Show that $pq = -4$

2

ii) M is the midpoint of PQ . Derive the locus of M and show that it is the

$$\text{parabola } y = \frac{x^2 + 8}{2}$$

2

iii) Find the focus of the parabola for M .

1

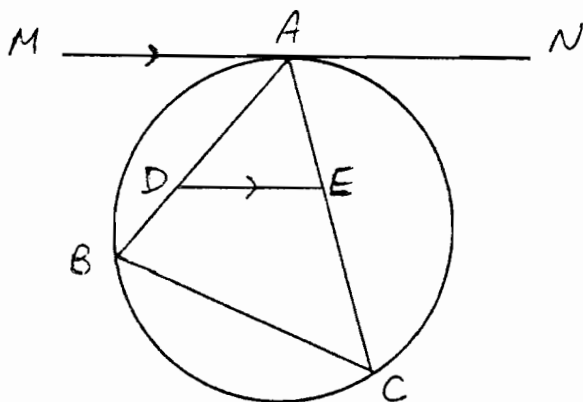
c) Prove by mathematical induction, that

$$1 \times 2^0 + 2 \times 2^1 + 3 \times 2^2 + \dots + n \times 2^{n-1} = 1 + (n-1)2^n \text{ where } n \text{ is a positive integer}$$

4

Question 5 (Start a new page)

a)



$\triangle ABC$ is inscribed in the circle.

MAN is tangent to the circle at

A and $DE \parallel MN$

- i) Copy the diagram onto your answer page
- ii) Prove that $BCED$ is a cyclic quadrilateral 3
- iii) Describe how to find the centre of the circle passing through B, C, E, D . 1

b) Given $f(x) = \frac{2}{x+1}$ for $x > -1$:

- i) Find the equation of the inverse function $f^{-1}(x)$ 1
- ii) On the same diagram, sketch the graphs of $y = f(x)$ and $y = f^{-1}(x)$. 3

Clearly show the coordinates of any points of intersection, intercepts on the coordinates axes and equations of any asymptotes.

- c) i) Sketch the curve $y = \sin^{-1}\left(\frac{x}{2}\right)$ 1
- ii) The area between the curve $y = \sin^{-1}\left(\frac{x}{2}\right)$ and the y axis is rotated about the y axis. 3

Find the volume thus generated.

Question 6 (Start a new page)

a) Differentiate $y = \tan^{-1}(\sin 3x)$ 2

b) In a population study, the population P of a town after t years is given by

$$P = 200 + Ae^{kt}.$$

The initial population was 300 and increased to 400 over 3 years.

i) Find the population after a further 2 years (nearest whole person) 3

ii) Find the rate of population growth after 10 years. 1

c) Kramer hits a golf ball from the top of the edge of a vertical cliff 25 metres above the sea. He hits it with an initial velocity of 50 m/s at a 30° angle of elevation.

The cliff top is taken as the point of origin.

i) Given $\ddot{x} = 0$ and $\ddot{y} = -10$, derive the equations of the horizontal and vertical components of the motion for the golf ball. 2

ii) Find the maximum height of the golf ball above the cliff. 2

iii) Find the angle at which the golf ball hits the water (nearest degree). 2

Question 7 (Start a new page)

a) A particle is moving according to the velocity equation $v = 4 - 2t$ m/s. Find the total distance it travels in the first 5 seconds of its motion. 2

b) A particle is moving with simple harmonic motion in a straight line with velocity

$$v^2 = 108 + 36x - 9x^2 \text{ where } x \text{ cm is its displacement from a point } O.$$

Initially it is at rest at $x = 6$ cm.

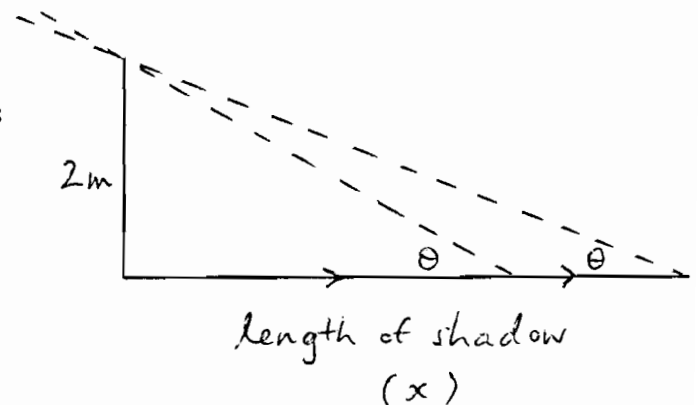
- | | | |
|------|---|---|
| i) | Use differentiation to find its acceleration in terms of x and find its maximum acceleration. | 2 |
| ii) | Find the maximum speed of the particle and the time when this first occurs. | 3 |
| iii) | Write an expression for the particle's displacement in terms of time t . | 1 |

c) A vertical pole, 2 metres high, casts a lengthening shadow as the sun sets.

4

At a particular instant, the shadow's length, x , is increasing by 0.3m/min.

Simultaneously, the angle of the Sun, θ , is decreasing by 0.05 radians/min.



Find the angle θ (to the nearest degree) when this is occurring.

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \quad \text{if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax \, dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax \, dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax \, dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax \, dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

2008 Extension Solutions

$$(1) a) i) \frac{dy}{dx} = 2x \cos^{-1} x - \frac{x^2}{\sqrt{1-x^2}}$$

$$ii) y = \frac{\log 3x}{\log 10}$$

$$\frac{dy}{dx} = \frac{1}{x \log 10}$$

$$b) P(3) = 1$$

$$\therefore 27 - 27 + 3p - 14 = 1$$

$$\therefore 3p = 15$$

$$\therefore p = 5$$

$$c) |x-3| < 4 \Rightarrow -4 < x-3 < 4$$
$$-1 < x < 7$$

$$|x-1| > 1 \Rightarrow x-1 > 1 \text{ or } x-1 < -1$$
$$x > 2 \text{ or } x < 0$$

$\therefore$ Simultaneous sol. is

$$-1 < x < 7 \text{ or } -1 < x < 0$$

$$d) 3 = \frac{-n + 5m}{m+n}$$

$$3m + 3n = -n + 5m$$

$$-2m = -4n$$

$$m = 2n$$

$$\frac{m}{n} = 2$$

$$\therefore m:n = 2:1$$

$$e) \frac{\sin^2 x}{2} + c$$

$$(2) a) \lim_{x \rightarrow \infty} \frac{3 - \frac{7}{x}}{\frac{5}{x^2} + 1} = \frac{3-0}{0+1}$$
$$= 3$$

$$b) \frac{dy}{dx} = 2x \Rightarrow m_1 = 6$$
$$m_2 = 5$$

$$\tan \theta = \left| \frac{6-5}{1+30} \right|$$
$$= \frac{1}{31}$$

$$\therefore \theta \approx 2^\circ$$

$$c) i) t = \frac{-2 \pm \sqrt{4+4}}{2}$$
$$= \frac{-2 \pm 2\sqrt{2}}{2}$$
$$= -1 \pm \sqrt{2}$$

$$ii) \tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}$$

$$\tan 45^\circ = \frac{2 \tan 22.5^\circ}{1 - \tan^2 22.5^\circ}$$

$$\therefore 1 - \tan^2 22.5^\circ = 2 \tan 22.5^\circ$$

$$\therefore \tan^2 22.5^\circ + 2 \tan 22.5^\circ - 1 = 0$$

$$\therefore \tan 22.5^\circ = -1 + \sqrt{2} (> 0)$$

(from i) above)

$$d) i) 3\cos x - 2\sin x = A \cos(x+\alpha)$$

$$(A = \sqrt{13})$$

$$= \sqrt{13} \cos(x+\alpha)$$

$$\therefore \frac{3}{\sqrt{13}} \cos x - \frac{2}{\sqrt{13}} \sin x = \cos(x+\alpha)$$

$$= \cos x \cos \alpha - \sin x \sin \alpha$$

$$\therefore \cos \alpha = \frac{3}{\sqrt{13}} \left\{ \begin{array}{l} \alpha = 33.7^\circ \\ \sin \alpha = \frac{2}{\sqrt{13}} \end{array} \right.$$

$$\therefore 3\cos x - 2\sin x = \sqrt{13} \cos(x+33.7^\circ)$$

ii) max. value of $3\cos x - 2\sin x$
 $= \text{max value of } \sqrt{13} \cos(x+33.7^\circ)$
 $= \sqrt{13}$, and this occurs
when $\cos(x+33.7^\circ) = 1$
 $\therefore x + 33.7^\circ = 360^\circ (\text{not } 0^\circ)$

$$\therefore x = 326.3^\circ$$

e) Let $\tan^{-1} x = \alpha$, $\tan^{-1} y = \beta$
 $x = \tan \alpha$, $y = \tan \beta$

$$\frac{\sqrt{1+x^2}}{x} \quad \frac{\sqrt{1+y^2}}{y}$$

$$\sin(\tan^{-1} x + \tan^{-1} y)$$

$$= \sin(\alpha + \beta)$$

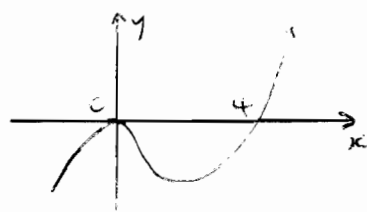
$$= \sin \alpha \cos \beta + \cos \alpha \sin \beta$$

$$= \frac{x}{\sqrt{1+x^2}} \cdot \frac{1}{\sqrt{1+y^2}} + \frac{1}{\sqrt{1+x^2}} \cdot \frac{y}{\sqrt{1+y^2}}$$

$$= \frac{x+y}{\sqrt{(1+x^2)(1+y^2)}}$$

3) a) $2\cos^2 \theta - 1 = \cos^2 \theta$
 $\cos^2 \theta = 1$
 $\cos \theta = \pm 1$
 $\theta = 0, \pi, 2\pi$

b) $\frac{x^2}{x-4} \times (x-4)^2 < 0$
 $x^2(x-4) < 0$



$$\therefore x < 0 \text{ or } 0 < x < 4$$

$$\text{or } x < 4 (x \neq 0)$$

c) $\int \frac{x}{x^2+4} dx + \int \frac{4}{x^2+4} dx$
 $= \frac{1}{2} \log(x^2+4) + 2 \tan^{-1}\left(\frac{x}{2}\right) + c$

d) $\int \frac{e^x}{\sqrt{1-4e^{2x}}} dx = \int \frac{u}{\sqrt{4-4u^2}} \frac{du}{u}$
 $u = e^x$
 $\frac{du}{dx} = e^x$
 $dx = \frac{du}{e^x}$
 $= \frac{du}{u}$

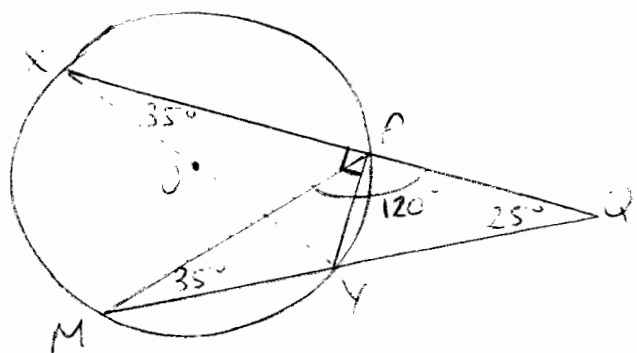
$$= \int \frac{1}{2\sqrt{4-u^2}} du$$

$$= \frac{1}{2} \sin^{-1}\left(\frac{u}{2}\right) + c$$

$$= \frac{1}{2} \sin^{-1}\left(\frac{e^x}{2}\right) + c$$

e) $(\alpha+\beta+\gamma)^2 = \alpha^2 + \beta^2 + \gamma^2 + 2\alpha\beta + 2\beta\gamma + 2\alpha\gamma$
 $\therefore \alpha^2 + \beta^2 + \gamma^2 = (\alpha+\beta+\gamma)^2 - 2(\alpha\beta + \beta\gamma + \alpha\gamma)$
 $= (6)^2 - 2(6)$

(4) a) i)



$\angle PMQ = 35^\circ$ (angles standing on same chord PY)

$\angle XPY = 90^\circ$ (angle in a semicircle)

$\angle YPQ = 120^\circ$ (angle sum at ΔMPQ)

$\therefore \angle MPY = 30^\circ$

b) i) $m_{OP} = \frac{p^2}{2p}$, $m_{OQ} = \frac{q^2}{2}$
 $= \frac{p}{2}$

$m_{OP} \times m_{OQ} = -1$ for perpend. lines

$\frac{p}{2} \times \frac{q}{2} = -1$

$\therefore pq = -4$ as reqd.

ii) M has coords $\left(\frac{2p+2q}{2}, \frac{p^2+q^2}{2} \right)$

$\therefore x = p+q$, $y = \frac{p^2+q^2}{2}$
 $= \frac{(p+q)^2 - 2pq}{2}$
 $= \frac{x^2 - 2(-4)}{2}$

iii) $2y = x^2 + 8$

$x^2 = 2y - 8$

$(x-0)^2 = 2(y-4)$

$\therefore$ vertex at $(0, 4)$ and $4a = 2$
 $\therefore a = \frac{1}{2}$

$\therefore$ focus at $(0, 4\frac{1}{2})$

c) Test $n=1 \Rightarrow LHS = 1 \times 2^0$, $RHS = 1 + 0 \times 2^1$
 $= 1$ $= 1$

$\therefore$ result is true for $n=1$

Assume result is true for $n=k$,

$\therefore$ assume that $S_k = 1 + (k-1)2^k$

Prove true for $n=k+1$,

$\therefore$ prove that $S_{k+1} = 1 + k \cdot 2^{k+1}$

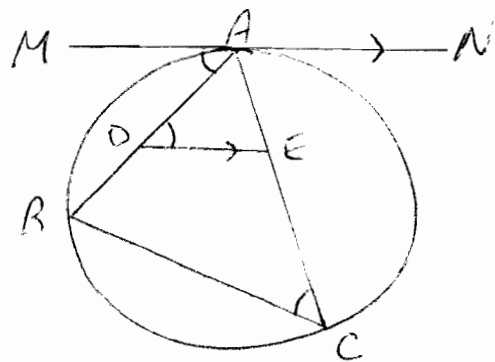
Now $S_{k+1} = S_k + T_{k+1}$
 $= 1 + (k-1)2^k + (k+1)2^k$
 $= 1 + 2^k(k-1+k+1)$
 $= 1 + 2^k \cdot 2k$
 $= 1 + k \cdot 2^k \cdot 2$
 $= 1 + k \cdot 2^{k+1}$

(shown)

So, if the result is true for $n=k$, then it has been proved true for $n=k+1$

Since the result is true for $n=1$, then from above it must be true for $n=1+1=2$ and so on for

⑤ a) i)



ii) $\angle MAN = \angle ADE$ (alt. angles
 $MN \parallel DE$)

$\angle MAN = \angle BCA$ (angle in alt.
 segment)

$\therefore \angle ADE = \angle BCA$

$\therefore BCED$ is a cyc. quadr. since
 exterior angle equals interior
 opposite angle.

iii) Perpendicular bisectors
 of at least 2 sides of $BCED$
 meet at the centre of the
 circle.

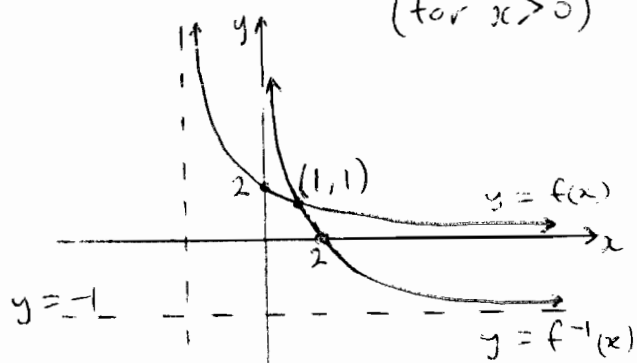
b) i) $x = \frac{2}{y+1}$

$$xy + x = 2$$

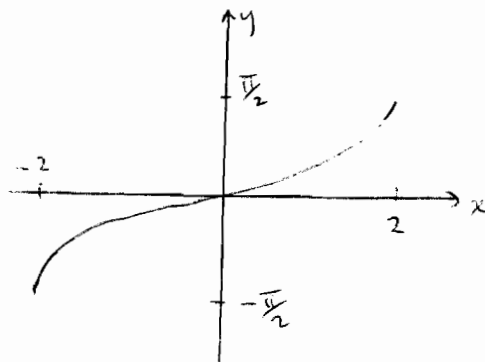
$$xy = 2 - x$$

$\therefore f^{-1}(x) \Rightarrow y = \frac{2-x}{x}$ or $\frac{2}{x} - 1$
 (for $x > 0$)

ii)



c) i)



ii) $V = 2\pi \int_0^{\pi/2} (2 \sin y)^2 dy$
 $= 8\pi \int_0^{\pi/2} \sin^2 y dy$
 $= 8\pi \int_0^{\pi/2} \frac{1}{2} (1 - \cos 2y) dy$
 $= 4\pi \left[y - \frac{\sin 2y}{2} \right]_0^{\pi/2}$
 $= 4\pi \left[\frac{\pi}{2} - 0 - (0 - 0) \right]$
 $= 2\pi^2$

⑥ a) $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$ ($u = \sin 3x$)
 $= \frac{1}{1+u^2} \times 3 \cos 3x$
 $= \frac{3 \cos 3x}{1 + \sin^2 3x}$

b) i) $P = 300, t = 0$:

$$300 = 200 + A \quad (A = 100)$$

$$\therefore P = 200 + 100e^{kt}$$

$P = 400, t = 3$

$$400 = 200 + 100e^{3k}$$

$$200 = 100e^{3k}$$

$$\therefore e^{3k} = 2$$

$$3k = \log 2$$

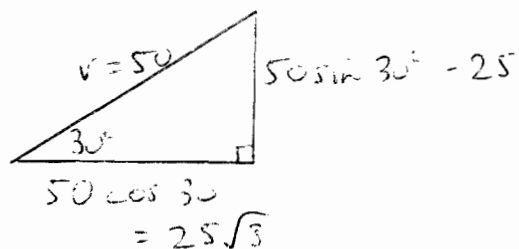
$$\therefore P = 200 + 100e^{\frac{t \log 2}{3}}$$

$$\text{When } t = 5, P = 200 + 100e^{\frac{5 \log 2}{3}} \\ \doteq 517 \text{ people}$$

$$\text{(ii)} \frac{dP}{dt} = 100e^{\frac{t \log 2}{3}} \times \frac{\log 2}{3}$$

$$\text{When } t = 10, \frac{dP}{dt} = 233 \text{ people per year}$$

(c)



i)

$$\ddot{x} = 0$$

$$\dot{x} = c$$

$$\text{When } t=0, \underline{\dot{x} = 25\sqrt{3}} \quad \text{When } t=0, y=25$$

$$\therefore \dot{x} = 25\sqrt{3}t + k \quad \therefore 25 = 0 + c \quad (c=25)$$

$$\text{When } t=0, x=0 \quad \therefore \underline{\underline{\dot{y} = -10t + 25}} \quad (k=0)$$

$$\therefore \underline{\underline{x = 25\sqrt{3}t}}$$

$$\therefore y = -5t^2 + 25t + k$$

$$\text{When } t=0, y=0$$

$$(k=0)$$

$$\therefore \underline{\underline{y = -5t^2 + 25t}}$$

$$\text{(ii) Max. height when } \dot{y} = 0 \text{ (i.e. } y = \frac{b}{2a})$$

$$\therefore -10t + 25 = 0$$

$$\therefore t = 2.5 \text{ seconds}$$

$$\text{(iii)} y = -25 \Rightarrow -25 = -5t^2 + 25t$$

$$5t^2 - 25t - 25 = 0$$

$$t^2 - 5t - 5 = 0$$

$$t = \frac{5 \pm \sqrt{25+20}}{2}$$

$$= \frac{5 \pm \sqrt{45}}{2}$$

$$= \frac{5 + 3\sqrt{5}}{2} \quad (t > 0)$$

$$\therefore \dot{y} = -10\left(\frac{5+3\sqrt{5}}{2}\right) + 25$$

$$\doteq -33.54 \quad \text{and } \dot{x} = 25\sqrt{3}$$

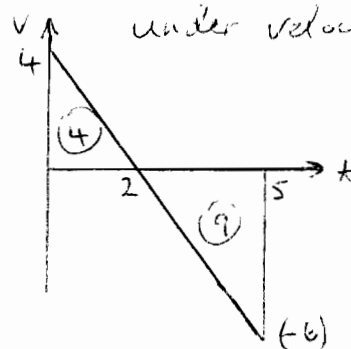


$$\tan \alpha = \frac{33.54}{25\sqrt{3}}$$

$$\therefore \alpha \doteq 38^\circ$$

⑦

a) dist travelled = total area under velocity graph.



$$\therefore \text{total dist. travelled} = 13 \text{ metres}$$

$$\text{b) i) } \dot{x} = \frac{dv}{dt} \left(\frac{1}{2} v^2 \right)$$

$$= \frac{d}{dx} \left(54 + 18x - \frac{9}{2} x^2 \right)$$

$$= 18 - 9x \text{ or } -9(x-2)$$

i) $V_{\max}$ when $x = \text{centre of oscillation}$
($x = 2$)

$$\therefore v^2 = 108 + 72 - 36$$

$$= 144$$

$$\therefore \text{max. speed} = 12 \text{ cm/s.}$$

$$\text{and time taken} = \frac{1}{4} \text{ of period}$$

$$= \frac{1}{4} \times \frac{2\pi}{3}$$

$$= \frac{\pi}{6} \text{ seconds}$$

iii) $x = b + a \cos(n\pi + \alpha)$

$$\therefore x = 2 + 4 \cos 3\pi$$

c) $\frac{dx}{dt} = 0.3, \frac{d\theta}{dt} = -0.05$

$$\tan \theta = \frac{2}{x}$$

$$\therefore x = \frac{2}{\tan \theta}$$

$$\frac{dx}{d\theta} = \frac{-2 \sec^2 \theta}{\tan^2 \theta}$$

$$= \frac{-2}{\sin^2 \theta}$$

$$\frac{dx}{d\theta} = \frac{dx}{dt} \times \frac{dt}{d\theta}$$

$$\frac{-2}{\sin^2 \theta} = 0.3 \times -0.05$$

$$= -0.015$$

$$\sin^2 \theta = \frac{1}{6}$$

$$\sin \theta = \frac{1}{\sqrt{6}} \quad (\theta \text{ is acute})$$

OR $\theta = \tan^{-1}\left(\frac{2}{x}\right)$

$$\frac{d\theta}{dx} = \frac{1}{1 + \left(\frac{2}{x}\right)^2} \times (-2x^{-2})$$

$$= \frac{1}{1 + \frac{4}{x^2}} \times \frac{-2}{x^2}$$

$$= \frac{-2}{x^2 + 4}$$

$$\frac{d\theta}{dx} = \frac{d\theta}{dt} \times \frac{dt}{dx}$$

$$\frac{-2}{x^2 + 4} = -0.05 \times \frac{1}{0.3}$$

$$= -\frac{1}{6}$$

$$12 = x^2 + 4$$

$$x^2 = 8$$

$$x = +\sqrt{8} \quad (x > 0)$$

$$\tan \theta = \frac{2}{\sqrt{8}}$$